

Title:

Exercise 10: Real-Time Streaming Analytics Using Hadoop MapReduce and Apache Cassandra

Hadoop + Cassandra Integration Exercise: Real-time & Batch Analytics

This comprehensive exercise demonstrates how to build a lambda architecture combining Hadoop (batch processing) and Cassandra (real-time data serving).

Exercise Overview

Use Case: Analyze song play data from a music streaming service to generate:

1. Real-time analytics (Cassandra for low-latency queries)
2. Batch analytics (Hadoop MapReduce for historical trends)
3. Hybrid recommendations (Combine real-time + historical data)

1.mapper_daily_aggregate.py

```
#!/usr/bin/env python3

import sys

for line in sys.stdin:
    line = line.strip()

    if not line:
        continue

    if line.startswith("user_id"):
        continue

    data = line.split(",")

    if len(data) < 6:
        continue

    date = data[1].split(" ")[0]
    user_id = data[0]
    song = data[2]
    artist = data[3]
    duration = data[5]
```

```
print(f"{date}\t{user_id},{song},{artist},{duration}")
```

2.reducer_daily_aggregate.py

```
#!/usr/bin/env python3
```

```
import sys
```

```
current_date = None
```

```
total_plays = 0
```

```
users = set()
```

```
total_duration = 0.0
```

```
song_count = {}
```

```
for line in sys.stdin:
```

```
    line = line.strip()
```

```
    if not line:
```

```
        continue
```

```
    date, value = line.split("\t")
```

```
    user_id, song, artist, duration = value.split(",")
```

```
    duration = float(duration)
```

```
    if current_date == date:
```

```
        total_plays += 1
```

```
        users.add(user_id)
```

```
        total_duration += duration
```

```
    key = (song, artist)
```

```
    song_count[key] = song_count.get(key, 0) + 1
```

```
else:
```

```
    if current_date is not None:
```

```
        avg = total_duration / total_plays
```

```
        top_song = max(song_count, key=song_count.get)
```

```
    print(
```

```
        f"{current_date}\t"
```

```
        f"{total_plays}\t"
```

```
        f"{len(users)}\t"
```

```

        f'{avg:.2f}\t'
        f'{top_song[0]}\t'
        f'{top_song[1]}"
    )

    current_date = date
    total_plays = 1
    users = {user_id}
    total_duration = duration
    song_count = {(song, artist): 1}

if current_date is not None:
    avg = total_duration / total_plays
    top_song = max(song_count, key=song_count.get)

print(
    f'{current_date}\t'
    f'{total_plays}\t'
    f'{len(users)}\t'
    f'{avg:.2f}\t'
    f'{top_song[0]}\t'
    f'{top_song[1]}"
)

```

3.mapper_top_songs.py

```

#!/usr/bin/env python3

import sys

for line in sys.stdin:
    line = line.strip()

    if not line:
        continue

    if line.startswith("user_id"):
        continue

    data = line.split(",")

    if len(data) < 6:
        continue

```

```
song = data[2]
artist = data[3]

print(f'{song}\t{artist}\t1')
```

4.reducer_top_songs.py

```
#!/usr/bin/env python3
```

```
import sys
```

```
current_song = None
current_artist = None
count = 0
```

```
for line in sys.stdin:
```

```
    line = line.strip()
```

```
    if not line:
        continue
```

```
    song, artist, value = line.split("\t")
```

```
    value = int(value)
```

```
    if current_song == song and current_artist == artist:
        count += value
```

```
    else:
        if current_song is not None:
            print(f'{current_song}\t{current_artist}\t{count}')
```

```
        current_song = song
        current_artist = artist
        count = value
```

```
    if current_song is not None:
        print(f'{current_song}\t{current_artist}\t{count}')
```

5.generate_batch_data.py

```
import csv
import random
```

```
from datetime import datetime, timedelta
```

```
songs = [  
    "Bohemian Rhapsody",  
    "Hotel California",  
    "Hey Jude",  
    "Stairway to Heaven",  
    "Comfortably Numb",  
    "Billie Jean",  
    "Back in Black",  
    "Smells Like Teen Spirit",  
    "Like a Prayer",  
    "With or Without You"  
]
```

```
artists = [  
    "Queen",  
    "The Beatles",  
    "Pink Floyd",  
    "Led Zeppelin",  
    "Michael Jackson",  
    "AC/DC",  
    "Madonna",  
    "U2",  
    "Nirvana",  
    "Eagles"  
]
```

```
start_date = datetime(2024, 1, 1)
```

```
with open("data/batch_plays.csv", "w", newline="") as file:
```

```
    writer = csv.writer(file)
```

```
    writer.writerow([  
        "user_id",  
        "timestamp",  
        "song",  
        "artist",  
        "session_id",  
        "duration"  
    ])
```

```
    for i in range(10000):
```

```

user = random.randint(1, 1000)

timestamp = start_date + timedelta(
    days=random.randint(0, 730),
    hours=random.randint(0, 23),
    minutes=random.randint(0, 59)
)

writer.writerow([
    user,
    timestamp.strftime("%Y-%m-%d %H:%M:%S"),
    random.choice(songs),
    random.choice(artists),
    random.randint(1000, 9999),
    round(random.uniform(2.5, 10.0), 2)
])

print("Dataset created successfully!")

```

6.cassandra_schema.cql

```

CREATE KEYSPACE IF NOT EXISTS streaming_analytics
WITH replication = {
'class':'SimpleStrategy',
'replication_factor':1
};

```

```

USE streaming_analytics;

```

```

CREATE TABLE user_listening_history (
    user_id INT,
    timestamp TIMESTAMP,
    song TEXT,
    artist TEXT,
    session_id INT,
    duration FLOAT,
    PRIMARY KEY(user_id, timestamp)
);

```

```

CREATE TABLE current_top_songs (
    date DATE,
    song TEXT,
    artist TEXT,

```

```
    play_count BIGINT,  
    PRIMARY KEY(date, song)  
);
```

```
CREATE TABLE hourly_stats (  
    hour TIMESTAMP PRIMARY KEY,  
    total_duration FLOAT,  
    total_plays INT,  
    unique_users INT  
);
```

```
CREATE TABLE user_sessions (  
    user_id INT,  
    session_id INT,  
    start_time TIMESTAMP,  
    end_time TIMESTAMP,  
    songs_count INT,  
    total_duration FLOAT,  
    PRIMARY KEY(user_id, session_id)  
);
```

```
CREATE TABLE daily_aggregates (  
    date DATE PRIMARY KEY,  
    total_plays INT,  
    unique_users INT,  
    avg_duration FLOAT,  
    top_song TEXT,  
    top_artist TEXT  
);
```

```
CREATE TABLE top_songs_daily (  
    date DATE,  
    rank INT,  
    song TEXT,  
    artist TEXT,  
    play_count BIGINT,  
    PRIMARY KEY(date, rank)  
);
```

```
CREATE TABLE user_recommendations (  
    user_id INT PRIMARY KEY,  
    recommended_song TEXT,  
    recommended_artist TEXT,  
    score FLOAT
```

);

```
Select sky@MS97377: ~/hadoop-cassandra-exercise
sky@MS97377:~$ docker --version
Docker version 29.6.1, build 8900f1d
sky@MS97377:~$ docker compose version
Docker Compose version v5.3.1
sky@MS97377:~$ docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS          NAMES
sky@MS97377:~$ mkdir -p ~/hadoop-cassandra-exercise/{data,scripts,output,jobs
sky@MS97377:~$ cd ~/hadoop-cassandra-exercise
sky@MS97377:~/hadoop-cassandra-exercise$ pwd
/home/sky/hadoop-cassandra-exercise
sky@MS97377:~/hadoop-cassandra-exercise$ nano docker-compose.yml
sky@MS97377:~/hadoop-cassandra-exercise$ docker compose up -d
[+] up 2/3
  Network... Created
  Contain... Creating
  service... Error response from daemon: Conflict. The container name "/cass
1e55771e6d30". You have to remove (or rename) that container to be able to re
Select sky@MS97377: ~/hadoop-cassandra-exercise
You have to remove (or rename) that container to be able to reuse that name.
sky@MS97377:~/hadoop-cassandra-exercise$ docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS          NAMES
0ba73b6b68f5   cassandra:4.0  "docker-entrypoint.s..." 2 weeks ago    Exited (143) 2 weeks ago    7000-7001/tcp, 0.0.0.0:9042->9042/tcp, [::]:9042->9042/tcp, 7199/t
4250a7eb2e07   ubuntu       "bash"                   2 weeks ago    Exited (0) 2 weeks ago
e4f402fbc66a   hello-world   "/hello"                 2 weeks ago    Exited (0) 2 weeks ago
sky@MS97377:~/hadoop-cassandra-exercise$ docker rm cassandra
cassandra
sky@MS97377:~/hadoop-cassandra-exercise$ docker stop cassandra
Error response from daemon: No such container: cassandra
sky@MS97377:~/hadoop-cassandra-exercise$ docker rm cassandra
Error response from daemon: No such container: cassandra
[+] up 1/17:~/hadoop-cassandra-exercise$ docker compose up -d
  Container cassandra Started
[+] up 1/17:~/hadoop-cassandra-exercise$ docker compose up -d
  Container cassandra Running
sky@MS97377:~/hadoop-cassandra-exercise$ docker ps -a
CONTAINER ID   IMAGE      COMMAND                  NAMES          CREATED        STATUS        PORTS
0fe50af8971f   cassandra:latest  "docker-entrypoint.s..."  cassandra      47 seconds ago  Up 45 seconds  7000-7001/tcp, 0.0.0.0:9042->9042/tcp, [::]:9042->9042/tcp, 7199/t
p, 0.0.0.0:9160->9160/tcp, [::]:9160->9160/tcp
4250a7eb2e07   ubuntu       "bash"                   happy_wing     2 weeks ago    Exited (0) 2 weeks ago
e4f402fbc66a   hello-world   "/hello"                 competent_curran 2 weeks ago    Exited (0) 2 weeks ago
[+] up 1/17:~/hadoop-cassandra-exercise$ docker rm cassandra
  Container cassandra Running
sky@MS97377:~/hadoop-cassandra-exercise$ docker compose up -d
sky@MS97377:~/hadoop-cassandra-exercise$ docker ps
CONTAINER ID   IMAGE      COMMAND                  NAMES          CREATED        STATUS        PORTS
0fe50af8971f   cassandra:latest  "docker-entrypoint.s..."  cassandra      About a minute ago  Up About a minute  7000-7001/tcp, 0.0.0.0:9042->9042/tcp, [::]:9042->9042/tcp, 7199/t
p, 0.0.0.0:9160->9160/tcp, [::]:9160->9160/tcp
sky@MS97377:~/hadoop-cassandra-exercise$ docker logs cassandra --tail 20
INFO [main] 2026-07-26T19:03:29.368 StorageService.java:3320 - Node /172.19.0.2:7000 state jump to NORMAL
INFO [main] 2026-07-26T19:03:29.372 StorageService.java:1466 - Auto-repair service is disabled via JVM property, skipping setup
INFO [main] 2026-07-26T19:03:29.373 Gossiper.java:2459 - Waiting for gossip to settle...
INFO [main] 2026-07-26T19:03:37.377 Gossiper.java:2490 - No gossip backlog; proceeding
INFO [main] 2026-07-26T19:03:37.574 CassandraDaemon.java:450 - Prewarming of auth caches is disabled
INFO [CompactionExecutor:3] 2026-07-26T19:03:37.628 CompactionTask.java:175 - Compacting (b5233f10-8924-11f1-9024-99a3257878d2) [/var/lib/cassandra/data/system/local-7ad54392bccdd35a684174e047860b377/nb-7-big-Data.db:level=0, /var/lib/cassandra/data/system/local-7ad54392bccdd35a684174e047860b377/nb-5-big-Data.db:level=0, /var/lib/cassandra/data/system/local-7ad54392bccdd35a684174e047860b377/nb-6-big-Data.db:level=0, /var/lib/cassandra/data/system/local-7ad54392bccdd35a684174e047860b377/nb-3-big-Data.db:level=0, /var/lib/cassandra/data/system/local-7ad54392bccdd35a684174e047860b377/nb-4-big-Data.db:level=0, /var/lib/cassandra/data/system/local-7ad54392bccdd35a684174e047860b377/nb-1-big-Data.db:level=0, /var/lib/cassandra/data/system/local-7ad54392bccdd35a684174e047860b377/nb-2-big-Data.db:level=0, ]
```

```
Select sky@MS97377: ~/hadoop-cassandra-exercise
b (repository: dirty), netty-handler-netty-handler-4.1.130.Final.41ff1eb (repository: dirty), netty-handler-proxy-netty-handler-proxy-4.1.130.Final.41ff1eb (repository:
dirty), netty-handler-ssl-ocsp-netty-handler-ssl-ocsp-4.1.130.Final.41ff1eb (repository: dirty), netty-resolver-netty-resolver-4.1.130.Final.41ff1eb (repository: dirty
), netty-transport-netty-transport-4.1.130.Final.41ff1eb (repository: dirty), netty-transport-classes-epoll-netty-transport-classes-epoll-4.1.130.Final.41ff1eb (reposit
ory: dirty), netty-transport-classes-kqueue-netty-transport-classes-kqueue-4.1.130.Final.41ff1eb (repository: dirty), netty-transport-native-epoll-netty-transport-nativ
e-epoll-4.1.130.Final.41ff1eb (repository: dirty), netty-transport-native-unix-common-netty-transport-native-unix-common-4.1.130.Final.41ff1eb (repository: dirty)]
INFO [main] 2026-07-26T19:03:37,736 PipelineConfigurator.java:148 - Starting listening for CQL clients on /0.0.0.0:9042 (unencrypted)...
INFO [main] 2026-07-26T19:03:37,787 CassandraDaemon.java:744 - Startup complete
INFO [CompactionExecutor:3] 2026-07-26T19:03:37,922 CompactionTask.java:274 - Compacted (b5233f10-8924-11f1-9024-99a3257878d2) 7 sstables to [/opt/cassandra/data/data/
system/local-7ad54392bccdd35a684174e047860b377/nb-8-big.] to level=0. 8718 to 6888 (~78% of original) in 286ms. Read Throughput = 2.967KiB/s, Write Throughput = 2.344K
iB/s, Row Throughput = ~2/s. 7 total partitions merged to 1. Partition merge counts were {7:1, }. Time spent writing keys = 123ms
INFO [NonPeriodicTasks:1] 2026-07-26T19:03:37,934 BigFormat.java:231 - Deleting sstable: /opt/cassandra/data/data/system/local-7ad54392bccdd35a684174e047860b377/nb-7-bi
g
INFO [NonPeriodicTasks:1] 2026-07-26T19:03:37,938 BigFormat.java:231 - Deleting sstable: /opt/cassandra/data/data/system/local-7ad54392bccdd35a684174e047860b377/nb-5-bi
g
INFO [NonPeriodicTasks:1] 2026-07-26T19:03:37,940 BigFormat.java:231 - Deleting sstable: /opt/cassandra/data/data/system/local-7ad54392bccdd35a684174e047860b377/nb-6-bi
g
INFO [NonPeriodicTasks:1] 2026-07-26T19:03:37,949 BigFormat.java:231 - Deleting sstable: /opt/cassandra/data/data/system/local-7ad54392bccdd35a684174e047860b377/nb-3-bi
g
INFO [NonPeriodicTasks:1] 2026-07-26T19:03:37,952 BigFormat.java:231 - Deleting sstable: /opt/cassandra/data/data/system/local-7ad54392bccdd35a684174e047860b377/nb-4-bi
g
INFO [NonPeriodicTasks:1] 2026-07-26T19:03:37,955 BigFormat.java:231 - Deleting sstable: /opt/cassandra/data/data/system/local-7ad54392bccdd35a684174e047860b377/nb-1-bi
g
INFO [NonPeriodicTasks:1] 2026-07-26T19:03:37,958 BigFormat.java:231 - Deleting sstable: /opt/cassandra/data/data/system/local-7ad54392bccdd35a684174e047860b377/nb-2-bi
g
INFO [OptionalTasks:1] 2026-07-26T19:03:47,824 CassandraRoleManager.java:454 - Created default superuser role 'cassandra'
sky@MS97377:~/hadoop-cassandra-exercise$ docker exec -it cassandra cqlsh
ATTENTION: All commands will be saved to history file: /root/.cassandra/cqlsh_history
This may include sensitive information such as passwords.
To disable history, use --disable-history or set 'disabled = true' in the [history] section of cqlshrc.
See https://cassandra.apache.org/doc/latest/tools/cqlsh.html for more information.

Connected to AnalyticsCluster at 127.0.0.1:9042
[cqlsh 6.2.0 | Cassandra 5.0.8 | CQL spec 3.4.7 | Native protocol v5]
Use HELP for help.
cqlsh> DESC KEYSPACES;

system          system_distributed  system_traces  system_virtual_schema
system_auth      system_schema        system_views

cqlsh> exit;
sky@MS97377:~/hadoop-cassandra-exercise$ docker exec -it cassandra cqlsh
ATTENTION: All commands will be saved to history file: /root/.cassandra/cqlsh_history
This may include sensitive information such as passwords.
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Use HELP for help.
cqlsh> DESC KEYSPACES;

system          system_distributed  system_traces  system_virtual_schema
system_auth      system_schema        system_views

cqlsh> exit;
sky@MS97377:~/hadoop-cassandra-exercise$ nano scripts/cassandra_schema.cql
sky@MS97377:~/hadoop-cassandra-exercise$ docker exec -i cassandra cqlsh -f /scripts/cassandra_schema.cql
/scripts/cassandra_schema.cql:18:InvalidRequest: Error from server: code=2200 [Invalid query] message="Cannot mix counter and non counter columns in the same table"

user_listening_history

sky@MS97377:~/hadoop-cassandra-exercise$ docker exec -it cassandra cqlsh
ATTENTION: All commands will be saved to history file: /root/.cassandra/cqlsh_history
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Connected to AnalyticsCluster at 127.0.0.1:9042
[cqlsh 6.2.0 | Cassandra 5.0.8 | CQL spec 3.4.7 | Native protocol v5]
Use HELP for help.
cqlsh> USE streaming_analytics;
cqlsh:streaming_analytics> DESC TABLES;

user_listening_history

cqlsh:streaming_analytics> DROP TABLE IF EXISTS current_top_songs;
cqlsh:streaming_analytics> exit;
sky@MS97377:~/hadoop-cassandra-exercise$ nano scripts/cassandra_schema.cql
sky@MS97377:~/hadoop-cassandra-exercise$ play_count BIGINT
play_count: command not found
sky@MS97377:~/hadoop-cassandra-exercise$ nano scripts/cassandra_schema.cql
```

```

Select sky@MS97377: ~/hadoop-cassandra-exercise
play_count: command not found
sky@MS97377:~/hadoop-cassandra-exercise$ nano scripts/cassandra_schema.cql
sky@MS97377:~/hadoop-cassandra-exercise$ docker exec -i cassandra cqlsh -f /scripts/cassandra_schema.cql

current_top_songs  user_listening_history

sky@MS97377:~/hadoop-cassandra-exercise$ docker exec -it cassandra cqlsh

ATTENTION: All commands will be saved to history file: /root/.cassandra/cqlsh_history
This may include sensitive information such as passwords.
To disable history, use --disable-history or set 'disabled = true' in the [history] section of cqlshrc.
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Connected to AnalyticsCluster at 127.0.0.1:9042
[cqlsh 6.2.0 | Cassandra 5.0.8 | CQL spec 3.4.7 | Native protocol v5]
Use HELP for help.
cqlsh> USE streaming_analytics;
cqlsh:streaming_analytics> DESC TABLES;

current_top_songs  user_listening_history

cqlsh:streaming_analytics> exit;
sky@MS97377:~/hadoop-cassandra-exercise$ nano scripts/cassandra_schema.cql
sky@MS97377:~/hadoop-cassandra-exercise$ docker exec -i cassandra cqlsh -f /scripts/cassandra_schema.cql

current_top_songs  user_listening_history

docker exec -it cassandra cqlsh
/scripts/cassandra_schema.cql:62:OperationTimeout: errors={'127.0.0.1:9042': 'Client request timeout. See Session.execute_async(timeout)'}, last_host=127.0.0.1:9042
[[A

sky@MS97377:~/hadoop-cassandra-exercise$ docker exec -i cassandra cqlsh -f /scripts/cassandra_schema.cql

current_top_songs  hourly_stats      user_listening_history  user_sessions
daily_aggregates   top_songs_daily         user_recommendations

sky@MS97377:~/hadoop-cassandra-exercise$ docker exec -it cassandra cqlsh

ATTENTION: All commands will be saved to history file: /root/.cassandra/cqlsh_history
This may include sensitive information such as passwords.
To disable history, use --disable-history or set 'disabled = true' in the [history] section of cqlshrc.
See https://cassandra.apache.org/doc/latest/tools/cqlsh.html for more information.

Connected to AnalyticsCluster at 127.0.0.1:9042
[cqlsh 6.2.0 | Cassandra 5.0.8 | CQL spec 3.4.7 | Native protocol v5]
Use HELP for help.
cqlsh> USE streaming_analytics;
cqlsh:streaming_analytics> DESC TABLES;

current_top_songs  hourly_stats      user_listening_history  user_sessions
daily_aggregates   top_songs_daily         user_recommendations

cqlsh:streaming_analytics> exit;
sky@MS97377:~/hadoop-cassandra-exercise$ nano scripts/load_realtime_data.cql
sky@MS97377:~/hadoop-cassandra-exercise$ docker exec -i cassandra cqlsh -f /scripts/load_realtime_data.cql
sky@MS97377:~/hadoop-cassandra-exercise$ docker exec -it cassandra cqlsh

ATTENTION: All commands will be saved to history file: /root/.cassandra/cqlsh_history
This may include sensitive information such as passwords.
To disable history, use --disable-history or set 'disabled = true' in the [history] section of cqlshrc.
See https://cassandra.apache.org/doc/latest/tools/cqlsh.html for more information.

Connected to AnalyticsCluster at 127.0.0.1:9042
[cqlsh 6.2.0 | Cassandra 5.0.8 | CQL spec 3.4.7 | Native protocol v5]
Use HELP for help.
cqlsh> USE streaming_analytics;
cqlsh:streaming_analytics> SELECT * FROM current_top_songs;

date      | song              | artist      | play_count
-----|-----|-----|-----
2026-07-26 | Bohemian Rhapsody | Queen      | 1
2026-07-26 | Comfortably Numb  | Pink Floyd | 1
2026-07-26 | Hey Jude          | The Beatles | 1
(3 rows)
cqlsh:streaming_analytics> SELECT * FROM user_listening_history;

user_id | timestamp              | artist      | duration | session_id | song
-----|-----|-----|-----|-----|-----
1001    | 2026-07-26 19:18:02.495000+0000 | Queen      | 5.55     | 5001       | Bohemian Rhapsody
1003    | 2026-07-26 19:18:02.576000+0000 | Led Zeppelin | 8.02     | 5003       | Stairway to Heaven
1002    | 2026-07-26 19:18:02.557000+0000 | The Beatles | 7.11     | 5002       | Hey Jude
(3 rows)
cqlsh:streaming_analytics> SELECT * FROM hourly_stats;

hour              | total_duration | total_plays | unique_users
-----|-----|-----|-----
2026-07-26 19:18:02.599000+0000 | 20.68 | 3 | 3
(1 rows)
cqlsh:streaming_analytics> SELECT * FROM user_sessions;

user_id | session_id | end_time              | songs_count | start_time              | total_duration
-----|-----|-----|-----|-----|-----
1001    | 5001       | 2026-07-26 19:18:02.607000+0000 | 1 | 2026-07-26 19:18:02.607000+0000 | 5.55
1002    | 5002       | 2026-07-26 19:18:02.613000+0000 | 1 | 2026-07-26 19:18:02.613000+0000 | 7.11
(2 rows)
cqlsh:streaming_analytics> exit;
sky@MS97377:~/hadoop-cassandra-exercise$ nano scripts/generate_batch_data.py
sky@MS97377:~/hadoop-cassandra-exercise$ for i in range(10000):
-bash: syntax error near unexpected token `('
sky@MS97377:~/hadoop-cassandra-exercise$ python3 scripts/generate_batch_data.py
Dataset created successfully!
Location: /home/sky/hadoop-cassandra-exercise/data/batch_plays.csv
sky@MS97377:~/hadoop-cassandra-exercise$ ls data
batch_plays.csv
sky@MS97377:~/hadoop-cassandra-exercise$ head data/batch_plays.csv
user_id,timestamp,song,artist,session_id,duration
623,2024-04-02 08:23:00,Stairway to Heaven,Eagles,2391,3.91
901,2024-08-10 19:57:00,Hotel California,Pink Floyd,7608,6.01
593,2025-12-28 17:06:00,Bohemian Rhapsody,The Beatles,2361,7.9
693,2025-06-13 13:59:00,Hey Jude,U2,4434,5.39
261,2025-07-02 20:37:00,Hey Jude,Eagles,1063,8.18
761,2025-12-12 21:41:00,Like a Prayer,Mirvana,7097,8.14
58,2025-07-12 21:30:00,Comfortably Numb,Madonna,4484,9.99
221,2025-05-07 00:07:00,Billie Jean,U2,6007,6.48
88,2025-12-14 10:49:00,Hey Jude,The Beatles,2943,9.23

```

```
sky@MS97377:~/hadoop-cassandra-exercise$ hdfs dfs -mkdir -p /user/$USER/analytics/input
sky@MS97377:~/hadoop-cassandra-exercise$ hdfs dfs -put -f data/batch_plays.csv /user/$USER/analytics/input/
sky@MS97377:~/hadoop-cassandra-exercise$ hdfs dfs -ls /user/$USER/analytics/input
```

```
hdfs dfs -cat /user/$USER/analytics/input/batch_plays.csv | head
^C
^XX
```

```
sky@MS97377:~/hadoop-cassandra-exercise$
sky@MS97377:~/hadoop-cassandra-exercise$
sky@MS97377:~/hadoop-cassandra-exercise$
sky@MS97377:~/hadoop-cassandra-exercise$ hdfs dfs -ls /user/$USER/analytics/input
Found 1 items
-rw-r--r-- 1 sky supergroup 602606 2026-07-27 00:54 /user/sky/analytics/input/batch_plays.csv
sky@MS97377:~/hadoop-cassandra-exercise$ hdfs dfs -cat /user/$USER/analytics/input/batch_plays.csv | head
user_id,timestamp,song,artist,session_id,duration
623,2024-04-02 08:23:00,Stairway to Heaven,Eagles,2391,3.91
901,2024-08-10 19:57:00,Hotel California,Pink Floyd,7608,6.01
593,2025-12-28 17:06:00,Bohemian Rhapsody,The Beatles,2361,7.9
693,2025-06-13 13:59:00,Hey Jude,U2,4434,5.39
261,2025-07-02 20:37:00,Hey Jude,Eagles,1963,8.18
761,2025-12-12 21:41:00,Like a Prayer,Nirvana,7097,8.14
58,2025-07-12 21:30:00,Comfortably Numb,Madonna,4484,9.99
221,2025-05-07 00:07:00,Billie Jean,U2,6007,6.48
88,2025-12-14 10:49:00,Hey Jude,The Beatles,2943,9.23
cat: Unable to write to output stream.
```

```
sky@MS97377:~/hadoop-cassandra-exercise$
```

```
sky@MS97377:~/hadoop-cassandra-exercise$
```

```
cat: Unable to write to output stream.
```

```
sky@MS97377:~/hadoop-cassandra-exercise$ nano scripts/mapper_daily_aggregate.py
sky@MS97377:~/hadoop-cassandra-exercise$ chmod +x scripts/mapper_daily_aggregate.py
sky@MS97377:~/hadoop-cassandra-exercise$ head data/batch_plays.csv | python3 scripts/mapper_daily_aggregate.py
2024-04-02 623,Stairway to Heaven,Eagles,3.91
2024-08-10 901,Hotel California,Pink Floyd,6.01
2025-12-28 593,Bohemian Rhapsody,The Beatles,7.9
2025-06-13 693,Hey Jude,U2,5.39
2025-07-02 261,Hey Jude,Eagles,8.18
2025-12-12 761,Like a Prayer,Nirvana,8.14
2025-07-12 58,Comfortably Numb,Madonna,9.99
2025-05-07 221,Billie Jean,U2,6.48
2025-12-14 88,Hey Jude,The Beatles,9.23
```

```
sky@MS97377:~/hadoop-cassandra-exercise$ nano scripts/reducer_daily_aggregate.py
sky@MS97377:~/hadoop-cassandra-exercise$ chmod +x scripts/reducer_daily_aggregate.py
sky@MS97377:~/hadoop-cassandra-exercise$ head -100 data/batch_plays.csv \
> | python3 scripts/mapper_daily_aggregate.py \
> | sort \
> | python3 scripts/reducer_daily_aggregate.py
```

```
2024-01-14 1 1 9.20 Bohemian Rhapsody Queen
2024-01-22 1 1 2.51 Like a Prayer Pink Floyd
2024-01-23 1 1 6.47 Smells Like Teen Spirit Led Zeppelin
2024-02-05 1 1 5.07 With or Without You Nirvana
2024-03-08 1 1 2.58 Bohemian Rhapsody Nirvana
2024-03-18 2 2 7.17 Back in Black Michael Jackson
2024-03-29 2 2 7.83 Hey Jude Nirvana
2024-03-30 1 1 8.80 Billie Jean Eagles
2024-04-02 1 1 3.91 Stairway to Heaven Eagles
2024-04-09 1 1 9.60 Smells Like Teen Spirit Michael Jackson
2024-04-24 1 1 9.94 Bohemian Rhapsody Madonna
2024-04-29 1 1 3.95 With or Without You Eagles
2024-05-03 1 1 6.84 Hey Jude Eagles
2024-05-04 1 1 9.90 With or Without You Pink Floyd
2024-05-09 1 1 9.24 Like a Prayer Eagles
2024-05-16 1 1 5.35 Back in Black The Beatles
2024-05-25 1 1 4.68 Like a Prayer Pink Floyd
2024-05-31 1 1 9.51 Smells Like Teen Spirit Eagles
2024-06-15 1 1 3.43 With or Without You Queen
2024-06-16 1 1 6.53 Comfortably Numb Pink Floyd
2024-06-29 1 1 8.07 Comfortably Numb U2
2024-07-22 1 1 4.13 Comfortably Numb U2
2024-07-29 1 1 9.49 Comfortably Numb Madonna
2024-08-06 2 2 7.56 Comfortably Numb U2
2024-08-10 2 2 5.90 Back in Black Queen
```

```
1596 NameNode
```

```
sky@MS97377:~/hadoop-cassandra-exercise$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
0fe50af8971f	cassandra:latest	"docker-entrypoint.s..."	16 hours ago	Up 2 minutes	7000-7001/tcp, 0.0.0.0:9042->9042/tcp, [::]:9042->9042/tcp, 7199/tcp, 0.0.0.0:9160->9160/tcp, [::]:9160->9160/tcp

```
sky@MS97377:~/hadoop-cassandra-exercise$ hdfs dfs -ls /user/$USER/analytics/output/daily_aggregates
```

```
Found 2 items
-rw-r--r-- 1 sky supergroup 0 2026-07-27 01:57 /user/sky/analytics/output/daily_aggregates/_SUCCESS
-rw-r--r-- 1 sky supergroup 34549 2026-07-27 01:57 /user/sky/analytics/output/daily_aggregates/part-00000
sky@MS97377:~/hadoop-cassandra-exercise$ hdfs dfs -cat /user/$USER/analytics/output/daily_aggregates/part-00000 | head -20
2024-01-01 11 11 6.02 Bohemian Rhapsody Led Zeppelin
2024-01-02 12 12 6.22 Bohemian Rhapsody Queen
2024-01-03 12 12 5.98 Back in Black U2
2024-01-04 12 12 6.41 Billie Jean Michael Jackson
2024-01-05 13 13 6.71 Smells Like Teen Spirit Led Zeppelin
2024-01-06 14 14 5.46 Comfortably Numb Pink Floyd
2024-01-07 13 13 6.54 Stairway to Heaven Eagles
2024-01-08 14 14 5.72 Comfortably Numb Madonna
2024-01-09 6 6 7.48 Billie Jean AC/DC
2024-01-10 19 19 5.84 Back in Black Queen
2024-01-11 12 12 5.70 Bohemian Rhapsody Eagles
2024-01-12 16 15 6.04 Smells Like Teen Spirit Nirvana
2024-01-13 9 9 6.37 Billie Jean Nirvana
2024-01-14 20 19 5.85 Bohemian Rhapsody Michael Jackson
2024-01-15 16 15 5.80 With or Without You AC/DC
2024-01-16 10 10 6.38 Stairway to Heaven AC/DC
2024-01-17 18 18 6.72 Smells Like Teen Spirit Led Zeppelin
2024-01-18 18 17 6.32 Like a Prayer Pink Floyd
2024-01-19 6 6 5.86 Hotel California Pink Floyd
2024-01-20 14 14 7.37 Back in Black Nirvana
```

```
cat: Unable to write to output stream.
```

```
sky@MS97377:~/hadoop-cassandra-exercise$ nano scripts/mapper_top_songs.py
sky@MS97377:~/hadoop-cassandra-exercise$ chmod +x scripts/mapper_top_songs.py
sky@MS97377:~/hadoop-cassandra-exercise$ head data/batch_plays.csv | python3 scripts/mapper_top_songs.py
Stairway to Heaven Eagles 1
Hotel California Pink Floyd 1
Bohemian Rhapsody The Beatles 1
Hey Jude U2 1
Hey Jude Eagles 1
Like a Prayer Nirvana 1
Comfortably Numb Madonna 1
```

```

sky@MS97377: ~/hadoop-cassandra-exercise
cat: Unable to write to output stream.
sky@MS97377:~/hadoop-cassandra-exercise$ nano scripts/mapper_top_songs.py
sky@MS97377:~/hadoop-cassandra-exercise$ chmod +x scripts/mapper_top_songs.py
sky@MS97377:~/hadoop-cassandra-exercise$ head data/batch_plays.csv | python3 scripts/mapper_top_songs.py
Stairway to Heaven      Eagles 1
Hotel California        Pink Floyd 1
Bohemian Rhapsody        The Beatles 1
Hey Jude                U2 1
Hey Jude                Eagles 1
Like a Prayer            Nirvana 1
Comfortably Numb         Madonna 1
Billie Jean              U2 1
Hey Jude                The Beatles 1
sky@MS97377:~/hadoop-cassandra-exercise$ nano scripts/reducer_top_songs.py
sky@MS97377:~/hadoop-cassandra-exercise$ chmod +x scripts/reducer_top_songs.py
sky@MS97377:~/hadoop-cassandra-exercise$ head -100 data/batch_plays.csv \
> | python3 scripts/mapper_top_songs.py \
sort \
| python> | sort \
> | python3 scripts/reducer_top_songs.py
Back in Black  AC/DC 1
Back in Black  Eagles 1
Back in Black  Led Zeppelin 1
Back in Black  Madonna 1
Back in Black  Michael Jackson 1
Back in Black  Queen 2
Back in Black  The Beatles 3
Billie Jean    AC/DC 1
Billie Jean    Eagles 2
Billie Jean    Led Zeppelin 1
Billie Jean    Michael Jackson 1
Billie Jean    Queen 2
Billie Jean    U2 1
Bohemian Rhapsody AC/DC 1
Bohemian Rhapsody Led Zeppelin 1
Bohemian Rhapsody Madonna 2
Bohemian Rhapsody Michael Jackson 1
Bohemian Rhapsody Nirvana 2
Bohemian Rhapsody Pink Floyd 3
Bohemian Rhapsody Queen 3
Bohemian Rhapsody The Beatles 2
Comfortably Numb Eagles 1
Comfortably Numb Madonna 2
Comfortably Numb Michael Jackson 4
Bohemian Rhapsody The Beatles 2
Comfortably Numb Eagles 1
Comfortably Numb Madonna 2
Comfortably Numb Michael Jackson 4
Comfortably Numb Queen 1
Comfortably Numb U2 3
Hey Jude        Eagles 3
Hey Jude        Led Zeppelin 1
Hey Jude        Nirvana 2
Hey Jude        Queen 1
Hey Jude        The Beatles 1
Hey Jude        U2 2
Hotel California Michael Jackson 1
Hotel California Pink Floyd 1
Hotel California Queen 1
Hotel California The Beatles 1
Like a Prayer    AC/DC 2
Like a Prayer    Eagles 2
Like a Prayer    Led Zeppelin 2
Like a Prayer    Nirvana 1
Like a Prayer    Pink Floyd 4
Like a Prayer    Queen 3
Smells Like Teen Spirit AC/DC 1
Smells Like Teen Spirit Eagles 1
Smells Like Teen Spirit Led Zeppelin 3
Smells Like Teen Spirit Michael Jackson 1
Smells Like Teen Spirit Pink Floyd 1
Smells Like Teen Spirit Queen 3
Smells Like Teen Spirit The Beatles 1
Stairway to Heaven Eagles 1
Stairway to Heaven Led Zeppelin 2
Stairway to Heaven Michael Jackson 1
Stairway to Heaven Pink Floyd 1
Stairway to Heaven Queen 1
With or Without You AC/DC 1
With or Without You Eagles 1
With or Without You Led Zeppelin 2
With or Without You Madonna 1
With or Without You Michael Jackson 1
With or Without You Nirvana 1
With or Without You Pink Floyd 1
With or Without You Queen 1
sky@MS97377:~/hadoop-cassandra-exercise$ hdfs dfs -rm -r -f /user/$USER/analytics/output/top_songs

```

```

With or Without You      Queen      1
sky@MS97377:~/hadoop-cassandra-exercise$ hdfs dfs -rm -r -f /user/$USER/analytics/output/top_songs
sky@MS97377:~/hadoop-cassandra-exercise$ hadoop jar $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar \
> -files scripts/mapper_top_songs.py,scripts/reducer_top_songs.py \
> -input /user/$USER/analytics/input/batch_plays.csv \
> -output /user/$USER/analytics/output/top_songs \
> -mapper "python3 mapper_top_songs.py" \
> -reducer "python3 reducer_top_songs.py"
packageJobJar: [/tmp/hadoop-unjar5515640077516873351/] [] /tmp/streamjob6512250668468424081.jar tmpDir=null
2026-07-27 17:05:26,226 INFO client.DefaultNoHARMFaloverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-07-27 17:05:30,592 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1785151827822_0001
2026-07-27 17:05:27,369 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/sky/.staging/job_1785151827822_0001
2026-07-27 17:05:29,682 INFO mapred.FileInputFormat: Total input files to process : 1
2026-07-27 17:05:29,825 INFO mapreduce.JobSubmitter: number of splits:2
2026-07-27 17:05:30,592 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1785151827822_0001
2026-07-27 17:05:30,592 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-07-27 17:05:31,213 INFO conf.Configuration: resource-types.xml not found
2026-07-27 17:05:31,215 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-07-27 17:05:31,889 INFO impl.YarnClientImpl: Submitted application application_1785151827822_0001
2026-07-27 17:05:32,006 INFO mapreduce.Job: The url to track the job: http://MS97377.localdomain:8088/proxy/application_1785151827822_0001/
2026-07-27 17:05:32,013 INFO mapreduce.Job: Running job: job_1785151827822_0001

2026-07-27 17:10:30,418 INFO mapreduce.Job: Job job_1785151827822_0001 running in uber mode : false
2026-07-27 17:10:30,421 INFO mapreduce.Job: map 0% reduce 0%
2026-07-27 17:10:59,154 INFO mapreduce.Job: map 50% reduce 0%
2026-07-27 17:11:00,368 INFO mapreduce.Job: map 100% reduce 0%
2026-07-27 17:11:37,775 INFO mapreduce.Job: map 100% reduce 67%
2026-07-27 17:11:41,317 INFO mapreduce.Job: map 100% reduce 100%
2026-07-27 17:11:44,630 INFO mapreduce.Job: Job job_1785151827822_0001 completed successfully
2026-07-27 17:11:46,997 INFO mapreduce.Job: Counters: 54
    File System Counters
      FILE: Number of bytes read=294710
      FILE: Number of bytes written=1427880
      FILE: Number of read operations=0
      FILE: Number of large read operations=0
      FILE: Number of write operations=0
      HDFS: Number of bytes read=606930
      HDFS: Number of bytes written=248895
      HDFS: Number of read operations=11
      HDFS: Number of large read operations=0

sky@MS97377:~/hadoop-cassandra-exercise$
WRONG REDUCE=0
File Input Format Counters
  Bytes Read=606702
File Output Format Counters
  Bytes Written=248895
2026-07-27 17:11:46,997 INFO streaming.StreamJob: Output directory: /user/sky/analytics/output/top_songs
sky@MS97377:~/hadoop-cassandra-exercise$
sky@MS97377:~/hadoop-cassandra-exercise$
sky@MS97377:~/hadoop-cassandra-exercise$
sky@MS97377:~/hadoop-cassandra-exercise$ hdfs dfs -ls /user/$USER/analytics/output/top_songs

Found 2 items
-rw-r--r-- 1 sky supergroup 0 2026-07-27 17:11 /user/sky/analytics/output/top_songs/_SUCCESS
-rw-r--r-- 1 sky supergroup 248895 2026-07-27 17:11 /user/sky/analytics/output/top_songs/part-00000
sky@MS97377:~/hadoop-cassandra-exercise$
sky@MS97377:~/hadoop-cassandra-exercise$ hdfs dfs -cat /user/$USER/analytics/output/top_songs/part-00000 | head -20
Back in Black      Led Zeppelin      1
Back in Black      Queen              1
Back in Black      Nirvana            1
Back in Black      U2                 1
Back in Black      Led Zeppelin      1
Back in Black      Michael Jackson   1
Back in Black      Madonna            1
Back in Black      Led Zeppelin      1
Back in Black      Eagles             1
Back in Black      Queen              1
Back in Black      Pink Floyd         1
Back in Black      Eagles             1
Back in Black      U2                 1
Back in Black      Queen              1
Back in Black      Nirvana            1
Back in Black      Queen              1
Back in Black      Madonna            1
Back in Black      Eagles             1
Back in Black      The Beatles        1
Back in Black      Nirvana            1
cat: Unable to write to output stream.
2026-07-27 17:56:30,307 INFO streaming.StreamJob: Output directory: /user/sky/analytics/output/top_songs
sky@MS97377:~/hadoop-cassandra-exercise$
sky@MS97377:~/hadoop-cassandra-exercise$ hdfs dfs -ls /user/$USER/analytics/output/top_songs

Found 2 items
-rw-r--r-- 1 sky supergroup 0 2026-07-27 17:56 /user/sky/analytics/output/top_songs/_SUCCESS
-rw-r--r-- 1 sky supergroup 248895 2026-07-27 17:56 /user/sky/analytics/output/top_songs/part-00000
sky@MS97377:~/hadoop-cassandra-exercise$ hdfs dfs -cat /user/$USER/analytics/output/top_songs/part-00000 | head -20
^C
^X
cat: Filesystem closed

sky@MS97377:~/hadoop-cassandra-exercise$
sky@MS97377:~/hadoop-cassandra-exercise$ hdfs dfs -cat /user/$USER/analytics/output/top_songs/part-00000 | head -20
Back in Black      Led Zeppelin      1
Back in Black      Queen              1
Back in Black      Nirvana            1
Back in Black      U2                 1
Back in Black      Led Zeppelin      1
Back in Black      Michael Jackson   1
Back in Black      Madonna            1
Back in Black      Led Zeppelin      1
Back in Black      Eagles             1
Back in Black      Queen              1
Back in Black      Pink Floyd         1
Back in Black      Eagles             1
Back in Black      U2                 1
Back in Black      Queen              1
Back in Black      Nirvana            1
Back in Black      Queen              1
Back in Black      Madonna            1
Back in Black      Eagles             1
Back in Black      The Beatles        1
Back in Black      Nirvana            1
cat: Unable to write to output stream.
sky@MS97377:~/hadoop-cassandra-exercise$ docker exec -it cassandra cqlsh

```

```

sky@MS97377: ~/hadoop-cassandra-exercise
cat: Unable to write to output stream.
sky@MS97377:~/hadoop-cassandra-exercise$ docker exec -it cassandra cqlsh
ATTENTION: All commands will be saved to history file: /root/.cassandra/cqlsh_history
This may include sensitive information such as passwords.
To disable history, use --disable-history or set 'disabled = true' in the [history] section of cqlshrc.
See https://cassandra.apache.org/doc/latest/tools/cqlsh.html for more information.

Connected to AnalyticsCluster at 127.0.0.1:9042
[cqlsh 6.2.0 | Cassandra 5.0.8 | CQL spec 3.4.7 | Native protocol v5]
Use HELP for help.
cqlsh> USE streaming_analytics;
cqlsh:streaming_analytics> INSERT INTO daily_aggregates
(date,total_plays,unique_user ... (date,total_plays,unique_users,avg_duration,top_song,top_artist)VALUES
('2024-01-01',11,11,6.02, ... VALUES
... ('2024-01-01',11,11,6.02,'Bohemian Rhapsody','Led Zeppelin');
1-02',12,12,6.22,'Bohemian Rhapsody','Queen');

INSERT INTO daily_aggregates
(date,total_plays,unique_users,avg_duration,top_song,top_artist)
VALUES
... ('2024-01-03',12,12,5.98,'Back in Black','U2');cqlsh:streaming_analytics>
cqlsh:streaming_analytics> INSERT INTO daily_aggregates
... (date,total_plays,unique_users,avg_duration,top_song,top_artist)
... VALUES
... ('2024-01-02',12,12,6.22,'Bohemian Rhapsody','Queen');

cqlsh:streaming_analytics>
cqlsh:streaming_analytics> INSERT INTO daily_aggregates
... (date,total_plays,unique_users,avg_duration,top_song,top_artist)
... VALUES
... ('2024-01-03',12,12,5.98,'Back in Black','U2');

cqlsh:streaming_analytics> INSERT INTO top_songs_daily
(date,rank,song,artist,play_count)
VALUES
('2024-01-01',1,'Bohemian Rhapsody','Led Zeppelin',11);
... (date,rank,song,artist,play_count)
... VALUES
... ('2024-01-01',1,'Bohemian Rhapsody','Led Zeppelin',11);
... ('2024-01-02',1,'Bohemian Rhapsody','Queen',12);

INSERT INTO top_songs_daily
(date,rank,cqlsh:streaming_analytics)
VALUES
... ('2024-01-01',1,'Bohemian Rhapsody','Led Zeppelin',11);
... ('2024-01-02',1,'Bohemian Rhapsody','Queen',12);

INSERT INTO top_songs_daily
(date,rank,cqlsh:streaming_analytics)
VALUES
... ('2024-01-01',1,'Bohemian Rhapsody','Led Zeppelin',11);
... ('2024-01-02',1,'Bohemian Rhapsody','Queen',12);

INSERT INTO top_songs_daily
(date,rank,cqlsh:streaming_analytics)
VALUES
... ('2024-01-03',1,'Back in Black','U2',12);

cqlsh:streaming_analytics> SELECT * FROM daily_aggregates;

+-----+-----+-----+-----+-----+-----+
| date | avg_duration | top_artist | top_song | total_plays | unique_users |
+-----+-----+-----+-----+-----+-----+
| 2024-01-01 | 6.02 | Led Zeppelin | Bohemian Rhapsody | 11 | 11 |
| 2024-01-03 | 5.98 | U2 | Back in Black | 12 | 12 |
| 2024-01-02 | 6.22 | Queen | Bohemian Rhapsody | 12 | 12 |
+-----+-----+-----+-----+-----+-----+
(3 rows)
cqlsh:streaming_analytics> SELECT * FROM top_songs_daily;

+-----+-----+-----+-----+-----+
| date | rank | artist | play_count | song |
+-----+-----+-----+-----+-----+
| 2024-01-01 | 1 | Led Zeppelin | 11 | Bohemian Rhapsody |
| 2024-01-03 | 1 | U2 | 12 | Back in Black |
| 2024-01-02 | 1 | Queen | 12 | Bohemian Rhapsody |
+-----+-----+-----+-----+-----+
(3 rows)
cqlsh:streaming_analytics> SELECT * FROM current_top_songs;

+-----+-----+-----+-----+
| date | song | artist | play_count |
+-----+-----+-----+-----+
| 2026-07-26 | Bohemian Rhapsody | Queen | 1 |
| 2026-07-26 | Comfortably Numb | Pink Floyd | 1 |
| 2026-07-26 | Hey Jude | The Beatles | 1 |
+-----+-----+-----+-----+

```

```
sky@MS97377: ~/hadoop-cassandra-exercise
2026-07-26 | Comfortably Numb | Pink Floyd | 1
2026-07-26 | Hey Jude | The Beatles | 1

(3 rows)
cqlsh:streaming_analytics> SELECT * FROM user_listening_history LIMIT 5;

user_id | timestamp | artist | duration | session_id | song
-----|-----|-----|-----|-----|-----
1001 | 2026-07-26 19:18:02.495000+0000 | Queen | 5.55 | 5001 | Bohemian Rhapsody
1003 | 2026-07-26 19:18:02.576000+0000 | Led Zeppelin | 8.02 | 5003 | Stairway to Heaven
1002 | 2026-07-26 19:18:02.557000+0000 | The Beatles | 7.11 | 5002 | Hey Jude

(3 rows)
cqlsh:streaming_analytics> SELECT * FROM daily_aggregates;

date | avg_duration | top_artist | top_song | total_plays | unique_users
-----|-----|-----|-----|-----|-----
2024-01-01 | 6.02 | Led Zeppelin | Bohemian Rhapsody | 11 | 11
2024-01-03 | 5.98 | U2 | Back in Black | 12 | 12
2024-01-02 | 6.22 | Queen | Bohemian Rhapsody | 12 | 12

(3 rows)
cqlsh:streaming_analytics> SELECT * FROM top_songs_daily;

date | rank | artist | play_count | song
-----|-----|-----|-----|-----
2024-01-01 | 1 | Led Zeppelin | 11 | Bohemian Rhapsody
2024-01-03 | 1 | U2 | 12 | Back in Black
2024-01-02 | 1 | Queen | 12 | Bohemian Rhapsody

(3 rows)
cqlsh:streaming_analytics> SELECT * FROM hourly_stats;

hour | total_duration | total_plays | unique_users
-----|-----|-----|-----
2026-07-26 19:18:02.599000+0000 | 20.68 | 3 | 3

(1 rows)
cqlsh:streaming_analytics> _
```

Result

The Hadoop Streaming jobs for **Daily Aggregation** and **Top Songs Analysis** were executed successfully on the music streaming dataset stored in HDFS. The processed results were generated using Python Mapper and Reducer programs and successfully stored in Apache Cassandra. The output was verified using HDFS commands and Cassandra CQL queries.

1. Data Modeling

Why do we use separate tables for real-time and batch data? How do they complement each other?

Separate tables are used because real-time data requires fast read/write operations, while batch data is processed periodically for historical analysis. Real-time tables provide the latest information, whereas batch tables store aggregated and long-term analytics. Together, they offer both instant insights and comprehensive historical reports.

2. Consistency Levels

What consistency level would you use for real-time vs batch reads? Why?

- **Real-time Reads:** ONE – provides low latency and faster responses.
- **Batch Reads:** QUORUM – ensures higher consistency and more accurate analytical results.

3. Performance

Compare the latency of real-time queries vs batch queries. When would you use each?

Real-time queries have very low latency and return results immediately, making them suitable for live dashboards and monitoring. Batch queries have higher latency because they process large datasets, making them ideal for reporting, trend analysis, and historical data processing.

4. Data Duplication

We store data in both HDFS and Cassandra. Is this efficient? Why or why not?

Yes, it is efficient because each system serves a different purpose. HDFS provides reliable distributed storage for large-scale batch processing, while Cassandra offers fast access to frequently used data. Although it increases storage usage, it significantly improves overall system performance and scalability.

5. Fault Tolerance

What happens if Cassandra goes down? How does the Hadoop batch process handle it?

If Cassandra goes down, new data cannot be written or queried until the database is restored. However, the Hadoop batch process continues processing data stored in HDFS. Once Cassandra becomes available again, the processed results can be loaded into the database.

6. Scaling

How would you scale this system for 1 billion daily events?

The system can be scaled by:

- Adding more Hadoop DataNodes for distributed storage and processing.
- Expanding the Cassandra cluster with additional nodes.
- Increasing the replication factor for fault tolerance.
- Partitioning data across multiple nodes.
- Using parallel MapReduce jobs to process large datasets efficiently.

Conclusion

This experiment successfully demonstrated the integration of **Hadoop MapReduce** and **Apache Cassandra** for processing and storing streaming data. Hadoop Streaming was used to perform distributed data processing, while Cassandra provided efficient storage and retrieval of the processed results. The experiment enhanced the understanding of big data processing, distributed computing, and NoSQL database management for real-time analytics applications.